

CSC621 Data Structures and Algorithm

Lab Exercise 6 - DLL

This lab is based on your previous tutorials where an array was used to store marks, IDs and grades. In this exercise, you are required to use a linked list of students instead of an array.

Question 1

1. Download DLL program from Moodle.
2. Create *Student* class that encapsulates *name*, *marks* and *grade*.
3. Create necessary constructors, accessors and modifiers for the *Student* class.
4. Change the datatype of *data* attribute of *Node* class to *Student*.
5. Update the constructors, accessors and modifiers of *Node* class where needed.
6. Implement the following menus in your main program:

Choose an option:

I - Insert

D - Display

S – Stop

where **Insert** is used to insert a student detail to the beginning of the list, **Display** is to show all the data from the list in a tabular format and **Stop** is to terminate the program.

7. Test your program by inserting and displaying three student details.